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Attack-resilient adaptive distributed neurodynamic approach for solving noncooperative games

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ABSTRACT

This paper considers noncooperative games operating under adversarial conditions, including both exploratory and causative attacks, where players' dynamics are subject to private inequality constraints and external disturbances. To address this challenge, we propose an adaptive distributed neurodynamic approach with continuous-time switching communication. Crucially, and in contrast to existing methods that often rely on fixed switching sequences—which can still pose a significant risk of information leakage over time—the communication topology in our framework is designed to switch randomly within a finite set of directed strongly connected graphs. This randomness provides a stronger defense against exploratory attacks like eavesdropping. Under mild conditions on the switching frequency, the convergence of the approach is rigorously proved using singular perturbation techniques, while integrated disturbance-observer-based control ensures effective rejection of external disturbances. To overcome the additional challenges posed by causative attacks and severe communication limitations, this paper further develops an attack-resilient adaptive distributed approach with discrete-time switching communication. This strategy guarantees exponential convergence to the NE even when the communication network is under attack and is rigorously designed to strictly exclude Zeno behavior. Comprehensive simulation results demonstrate that the proposed approaches achieve faster convergence, stronger resilience against both types of attacks, and higher adaptability in dynamic environments compared to existing methods. The practical effectiveness and utility of the approaches are conclusively validated through their application to a self-organizing autonomous vehicle network, confirming their suitability for real-world noncooperative game scenarios.

1. Introduction

The Nash equilibrium seeking approaches for noncooperative games have attracted wide attention (Bahrami & Parniani, 2014; Cao et al., 2012; He et al., 2014; Huang et al., 2020; Li et al., 2018; Liu et al., 2022; Ma et al., 2016; Stankovic et al., 2012; Yang et al., 2015). From the noncooperative game-theoretic perspective, each player is described as a self-interested individual competing with others to reach the stable status, known as the "Nash equilibrium (NE)" (Nash, 1951). For example, in the plug-in electric vehicle charging control problem in a smart grid (Li et al., 2018), each vehicle acts as a selfish player aiming to minimize its own charging cost by adjusting its charging strategy. In this context, the energy consumption behavior of the plug-in electric vehicles can be modeled as a noncooperative game. Under such a scenario, the NE emerges as the equilibrium solution collectively adopted by all vehicles. Compared to centralized approaches (Cao et al., 2012; Yang et al., 2015), the distributed NE seeking approach, which does not re-

quire a global coordinator or a complete communication graph, is a crucial research problem in game-theoretic contexts. This inherent benefit of not depending on centralized information infrastructures has drawn a lot of research attention, making the problem an essential theoretical underpinning with wide-ranging applications in a number of engineering domains, such as the above mentioned distributed charging control for plug-in electric vehicles in the smart grids (Li et al., 2018) and mobile sensor network control for the velocity/force-actuated vehicles and unicycles (Stankovic et al., 2012). Recently, there has been a wealth of research on distributed NE seeking in noncooperative games (Gadjov & Pavel, 2019; Liu & Yi, 2023; Salehisadaghiani & Pavel, 2016; Ye et al., 2023; Ye & Hu, 2017; Zhang et al., 2021). In Salehisadaghiani and Pavel (2016), the authors pointed out that some practical engineering systems did not allow full communication with other systems, like multi-agent systems and self-organizing (ad hoc) networks. To address the consensus on actions' estimations in the scenarios with partial decision information, existing distributed approaches can be broadly categorized into

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several methodological frameworks. The passivity-based consensus protocol (Gadjov & Pavel, 2019) and the leader-following consensus protocol (Ye & Hu, 2017) represent two main streams for handling action estimation under partial decision information. Between the two protocols, the leader-following consensus protocol demonstrates significant advantages in terms of its simple structure and excellent extensibility due to its “separation” feature, which means that in the algorithms, communication-related issues are confined to the consensus module, and optimization-related issues are handled in the optimization module (Ye et al., 2023). This greatly enhances its extensibility and simplifies the design and analysis of its extensions (Chen et al., 2025; Feng et al., 2025; Huang et al., 2025; Liu & Yi, 2023; Tian et al., 2025; Wang et al., 2025; Ye et al., 2023; Zhang et al., 2021).

The aforementioned works primarily assume players compete in deterministic environments, which is an idealized scenario. However, real-world applications must account for various uncertainties, a challenge that has garnered significant research attention across different domains (Cai et al., 2023; Chen et al., 2024; Deng & Nian, 2018; He & Huang, 2024; Romano & Pavel, 2020). A prominent source of uncertainty is external disturbances (Cai et al., 2023), originating from environmental factors like wind, electromagnetic interference, or communication uncertainties (Chen et al., 2024; Deng & Nian, 2018; He & Huang, 2024; Romano & Pavel, 2020). When the dynamics of players are affected by such disturbances, actual actions may not be available. This severely impacts the effectiveness of approaches and the accuracy of NE seeking. Deng and Nian (2018) introduced the internal model technique to attenuate the exogenous disturbances and achieved distributed NE seeking for an aggregative game in a weight-balanced directed graph. However, as a classical class of passive disturbance rejection control, it was shown in Chen et al. (2016) that the internal model technique struggled to respond directly and quickly under strong disturbances. To overcome this limitation, Chen et al. (2024) designed the disturbance-observer-based control approach to achieve disturbance rejection. Moreover, due to limitations imposed by the engineering environment for ensuring production efficiency and safety, such as transmit power constraints in wireless networks (Katwe et al., 2023), the players are often subject to various constraints (Chai et al., 2024; Li et al., 2015; Liu et al., 2024; Lu & Zhu, 2022; Wang et al., 2024b, 2023; Zeng et al., 2025). Wang et al. (2024b) investigated an aggregative game problem where players' dynamics were subject to private and coupling constraints. Under a non-cooperative game with heterogeneous individual dynamics, Lu and Zhu (2022) proposed a distributed NE seeking algorithm based on adaptive gains, in which players' dynamics were subject to both local strategy set constraints and global nonlinear inequality constraints. Furthermore, research on NE seeking neurodynamic approaches that address both external disturbances and private inequality constraints in players' dynamics is crucial for enhancing the real-world applicability of these methods.

Communication attacks, including exploratory attack and causative attack (Ding & Yan, 2020), are another crucial aspect considered in real-world application environments. While most of the studies have focused on fixed communication topologies (Sun & Liu, 2018; Wang et al., 2024a; Yu et al., 2024), practical scenarios often involve vulnerabilities to exploratory attacks, such as Man-in-the-Middle (MITM) attacks. When an exploratory attack occurs, the attacker may attempt to extract available information, which can lead to the leakage of sensitive transmitted data. However, the switching communication mechanism could effectively prevent prolonged information leakage by dynamically changing the communication topology. Thus, considering switching communication topologies for privacy protection is essential, and the related issues within NE seeking field have garnered increasing attention (Chen et al., 2024; Ding et al., 2024b; He & Huang, 2024; Ye & Hu, 2018, 2021). For instance, Ding et al. (2024b) proposed a distributed event-triggered NE seeking algorithm, where the communication topology alternates between directed strongly connected and non-strongly connected graphs. Beyond privacy concerns, communication reliability under malicious attacks presents another major challenge. Causative attacks, such as delib-

erate link failures or Denial of Service (DoS) attacks, can disrupt communication channels and degrade system performance (Chen et al., 2025; Feng et al., 2025; Meng et al., 2024; Wang et al., 2022; Ye & Hu, 2018). When a causative attack occurs, the attacker may change the condition of communication, such as link failures as discussed in Ye and Hu (2018). Under a fixed communication topology subject to attacks and communication delays, Wang et al. (2022) proposed a distributed robust NE seeking algorithm in a noncooperative game. Further, Ye and Hu (2018) investigated distributed NE seeking in multi-agent games under switching topologies, in which communication is unavailable during specific time slots. Denial of service (DoS) as one of the causative attacks was considered in Meng et al. (2024), where the authors proposed a two-stage distributed NE seeking strategy to achieve attack-resilience. In a communication network with a large number of self-organizing players, due to physical layer constraints and security requirements for communication, the cost of communication is much higher than the cost of computation (Yi et al., 2022). To balance the communication and computation complexity, the communication mechanism was further considered. A dynamic event-triggered mechanism is designed with the switching-based compensation communication mechanism presented in Ding et al. (2024b), achieving discrete-time communication and effectively reducing the frequency of communication. However, research on NE seeking neurodynamic approach with switching communication under both exploratory attacks and causative attacks is rarely investigated. Due to the complexity of communication, this research has significant challenges.

Despite these advances, current literature exhibits two major limitations: (a) Most works address either disturbances or constraints separately, with limited research on integrated solutions; (b) Comprehensive frameworks handling both exploratory and causative attacks in switching communication environments remain under-explored. The contributions of this paper are summarized as follows:

- 1) A novel adaptive distributed approach is proposed for more general switching communication topologies, incorporating private inequality constraints and external disturbances. Unlike the fixed switching intervals in Chen et al. (2024), Ding et al. (2024b), He and Huang (2024), Ye and Hu (2021) and Ye and Hu (2018)—which can still pose a significant risk of information leakage over time—the switching intervals are unfixed and the instants are random, thereby offering stronger privacy protection against exploratory attacks. Furthermore, disturbance-observer-based control and an adaptive penalty method are introduced to reject disturbances and reduce dimensionality without adding auxiliary variables.
- 2) Causative attacks in communication are considered, and an attack-resilient adaptive distributed approach with discrete-time switching communication is developed. Differing from the attacks in Ye and Hu (2018), Wang et al. (2022), and Meng et al. (2024), this work addresses non-sustained causative attacks launched by an energy-limited attacker, where sustained attacks become a special case. Expanding the framework in Ye and Hu (2018), the impact of one attack may span one or more switching instants. Moreover, under a given switching frequency, the relationship between attack frequency and communication recovery time is established. The communication mechanism is proven to be Zeno-free.
- 3) In the self-organizing network of autonomous vehicles, the proposed adaptive distributed approaches are respectively applied in two different communication scenarios. Results show that under communication attacks, the vehicles balance target positioning with privacy protection, demonstrating the approaches' practical applicability and relevance in real-world noncooperative game settings.

The rest of this paper is organized as follows. In Section 2, some related preliminaries are introduced and the noncooperative game with private inequality constraints and external disturbances is formulated. In Section 3, an adaptive distributed approach with continuous-time switching communication is proposed. In Section 4, considering the communication under causative attack, an adaptive distributed ap-

proach with discrete-time switching communication is proposed. An example of an autonomous vehicle problem is provided in Section 5. Finally, conclusions are presented in Section 6.

2. Preliminaries and problem formulation

2.1. Preliminaries

Notations : $\mathbb{R}$, $\mathbb{R}^m$ and $\mathbb{R}^{m \times n}$ represent the sets of real numbers, m -dimensional vector and $m \times n$ real matrices, respectively. The transpose and the Euclidean norm of a vector $x \in \mathbb{R}^m$ are denoted by $x^\top$ and $\|x\|$, respectively. For a matrix B , its transpose and the Euclidean norm are $B^\top$ and $\|B\|$, respectively. Furthermore, $\otimes$ is the Kronecker product. $\text{col}(x_1, x_2, \dots, x_N) = [x_1^\top, x_2^\top, \dots, x_N^\top]^\top$ with $x_i \in \mathbb{R}^m$ and $i \in \{1, 2, \dots, N\}$. $[b_i]_{\text{vec}}$ ($\text{diag}\{b_i\}$) denotes a column vector (diagonal matrix) whose (principal diagonal) element is b_i . Respectively, identity matrix, zero matrix, and one matrix are denoted by I , $\mathbf{0}$ and $\mathbf{1}$ with each appropriate dimension. Denote $\max_{i \in \{1, \dots, N\}} \{b_i\}$ and $\min_{i \in \{1, \dots, N\}} \{b_i\}$ as the maximum and minimum values of multiple real constants b_i , respectively.

Graph theory : Some definitions related to graph theory are introduced briefly from [Godsil and Royle \(2001\)](#). A directed graph without self-loops is denoted by $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ with the node set $\mathcal{V} = \{1, 2, \dots, N\}$ and the edge set $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$. $\mathcal{N}_i = \{j : (j, i) \in \mathcal{E}\}$ is the neighbor set of node i , which means that node i can receive information from node j . A directed path consists of a sequence of connected edges. If there is a directed path between any pair of nodes, then the directed graph $\mathcal{G}$ is strongly connected. $A = [a_{ij}] \in \mathbb{R}^{N \times N}$ is the adjacency matrix, where $a_{ii} = 0$, $a_{ij} = 1$ if $(j, i) \in \mathcal{E}$, otherwise $a_{ij} = 0$. $D = \text{diag}\{d_1, d_2, \dots, d_N\} \in \mathbb{R}^{N \times N}$ is the degree matrix, where $d_i = \sum_{j=1}^N a_{ij}$ represents i th node's degree. The Laplacian matrix of $\mathcal{G}$ is denoted by $L = D - A = [l_{ij}] \in \mathbb{R}^{N \times N}$, where $l_{ij} = -a_{ij}$ if $i \neq j$, and $l_{ii} = d_i$.

In this paper, the communication topology is switching in a set of directed strongly connected graphs denoted by $\mathcal{C}_c = \{\mathcal{G}_1, \dots, \mathcal{G}_r\}$. Let $\mathcal{M}_c = \{1, 2, \dots, r\}$ represent the index set of the switching graphs. Denote a piecewise switching signal as $\delta(t) : \mathbb{R} \rightarrow \mathcal{M}_c$. Let t_p , $p = 1, 2, \dots, \infty$ be the switching time sequence, which means for $t \in [t_p, t_{p+1})$, $\mathcal{G}_{\delta(t)}$ remains unchanged. The Laplacian matrix of $\mathcal{G}_{\delta(t)}$ is defined correspondingly as $L^{\delta(t)} = D^{\delta(t)} - A^{\delta(t)}$, where $D^{\delta(t)}$, $A^{\delta(t)}$ are the degree matrix and adjacency matrix of $\mathcal{G}_{\delta(t)}$, respectively.

Remark 1. To mitigate the impact of exploratory attacks, such as Man-in-the-Middle (MITM) attacks, a random switching strategy for the communication topology is adopted. Specifically, both the dwell time (i.e., the interval $\Delta t_p = t_{p+1} - t_p$) and the switching sequence are designed to be random, rather than fixed or periodic. This randomized mechanism, implemented in Section 5, significantly enhances privacy protection by reducing the predictability of communication patterns, thereby making it more difficult for attackers to intercept or infer sensitive information continuously.

2.2. Problem formulation

Consider an N -players noncooperative game G with private inequality constraints.

$$\min_{x_i \in \mathcal{X}_i} f_i(x_i, x_{-i}) \quad (1)$$

where $x_i \in \mathbb{R}^m$ denotes the action of player i , $x_{-i} = \text{col}(x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_N)$, $x = (x_i, x_{-i})$, $\mathcal{X}_i = \{x_i \in \mathbb{R}^m | g_i(x_i) \leq 0\}$ and the overall feasible action space $\mathcal{X} = \mathcal{X}_1 \times \dots \times \mathcal{X}_N$. $f_i : \mathbb{R}^m \rightarrow \mathbb{R}$ and $g_i : \mathbb{R}^m \rightarrow \mathbb{R}$ are player i 's cost function and private constraint function, respectively.

Assume that the functions $f_i(x_i, x_{-i})$ and $g_i(x_i)$ for each $i \in \mathcal{V}$ are twice continuously differentiable, and $g_i(x_i)$ is convex with respect to x_i , i.e., for any $x_i, x'_i \in \mathbb{R}^m$, $\lambda \in [0, 1]$, there is $g_i(\lambda x_i + (1 - \lambda)x'_i) \leq \lambda g_i(x_i) + (1 - \lambda)g_i(x'_i)$.

Definition 1. ([Facchinei & Pang, 2009](#)) A NE of game (1) is an action profile $x^* = (x_i^*, x_{-i}^*)$ satisfying that for $x_i \in \mathcal{X}_i$,

$$f_i(x_i^*, x_{-i}^*) \leq f_i(x_i, x_{-i}^*), \quad \forall i \in \mathcal{V},$$

which means no player can decrease its cost by only changing its own action.

Denote the pseudo-gradient $\nabla_i f_i(x) = \frac{\partial f_i(x)}{\partial x_i}$ and the pseudo-gradient vector $F(x) = [\nabla_i f_i(x)]_{\text{vec}}$.

Assumption 1. For each $i \in \mathcal{V}$, $\nabla_i f_i(x)$ is globally l_i -Lipschitz continuous, i.e., there exists a constant $l_i > 0$ such that for any $x, x' \in \mathbb{R}^m$, $\|\nabla_i f_i(x) - \nabla_i f_i(x')\| \leq l_i \|x - x'\|$.

Assumption 2. $F(x)$ is q -strongly monotone, i.e., there exists a constant $q > 0$ such that for any $x, x' \in \mathbb{R}^m$, $z \in F(x)$ and $z' \in F(x')$, $(x - x')^\top (z - z') \geq q \|x - x'\|^2$.

Assumption 1 ensures that each player has a continuous gradient, which validates the gradient updates used in our gradient play framework and restricts the changes in gradient values with respect to different action values to facilitate the convergence analysis ([Ding et al., 2024a](#)). And **Assumption 2** indicates the strong convexity of the cost function f_i for $i \in \mathcal{V}$. These two assumptions guarantee the existence and uniqueness of NE ([Rosen, 1965](#), Theorem 2), which have been widely used in [Chen et al. \(2024\)](#), [Deng and Nian \(2018\)](#), [Ding et al. \(2024b\)](#) and [Gadjov and Pavel \(2019\)](#).

3. Adaptive distributed neurodynamic approach with continuous-time switching communication

In this section, an adaptive distributed NE seeking neurodynamic approach with continuous-time switching communication is proposed, which only considers exploratory attacks. And the convergence of the proposed approach is analyzed based on the singular perturbation technique.

3.1. Adaptive distributed approach with continuous-time communication

Consider that players have the following dynamics to adjust their own actions to minimize their own cost functions.

$$\dot{x}_i(t) = u_i(t) + d_i(t), \quad (2)$$

where $u_i(t) \in \mathbb{R}^m$ is control input, $d_i(t) \in \mathbb{R}^m$ is external disturbance determined by

$$d_i(t) = S d_i(t), \quad (3)$$

herein $S \subseteq \mathbb{R}^{m \times m}$ satisfies the following assumption.

Assumption 3. All the m eigenvalues of S are distinct and purely imaginary.

Assumption 3 can characterize various types of disturbances encountered in engineering applications such as unknown constant and harmonics with unknown phase and magnitude, and is frequently utilized in disturbance attenuation ([Chen et al., 2016, 2024](#)).

As is known to us, centralized approaches require each player to have direct access to strong coupling information x for computing its cost function $f_i(x)$. This necessitates either a centralized coordinator or a complete communication graph, imposing huge communication burdens, especially in games with a large number of players. To address the issue, the leader-following consensus protocol proposed in [Ye and Hu \(2017\)](#) is introduced in this paper, achieving NE seeking in a distributed manner.

$$\varepsilon \dot{y}_{ij}(t) = - \left(\sum_{r=1}^N a_{ir}^{\delta(t)} (y_{ij}(t) - y_{rj}(t)) - d_{ij}^{\delta(t)} (y_{ij}(t) - x_j(t)) \right), \quad (4)$$

where y_{ij} denotes the estimation of player i on player j 's action, $a_{ij}^{\delta(t)}$ is the element (i, j) of the adjacency matrix $A^{\delta(t)}$ where $\delta(t)$ is defined in Section 1, and $\varepsilon > 0$ is a singular perturbation parameter.

By the adaptive penalty method, the NE seeking of game (1) is transformed into solving the following problem

$$\min_{x_i \in \mathbb{R}^m} f_i(x_i, x_{-i}) + \sigma_i(t) G_i(x_i), \quad (5)$$

where $G_i(x_i) = \max\{0, g_i(x_i)\}$, $\sigma_i(t)$ is adaptive penalty parameter with $\sigma_i(0) > 0$ and satisfies

$$\dot{\sigma}_i(t) = G_i(x_i). \quad (6)$$

Due to the convexity of $g_i(x_i)$, the function $G_i(x_i)$ is convex. According to Theorem 1 in Sun and Hu (2021), there exists $\sigma_i^* > 0$ and when $\sigma_i > \sigma_i^*$, the proposed penalty method can derive players' actions to enter the constraints. Compared with Sun and Hu (2021), the adaptive penalty method can avoid the additional calculation of the lower bound σ^* . Then, the unique NE of game (1) x^* and σ_i^* satisfy $\mathbf{0}_{Nm} \in [\nabla f_i(x^*) + \sigma_i^* \partial G_i(x_i^*)]_{vec}$, where $\partial G_i(x_i^*)$ is the subgradient of $G_i(x_i^*)$.

Obviously, $d_i, i \in \mathcal{V}$ are non-vanishing external disturbances. For attenuating the disturbances to achieve disturbance rejection, the disturbance-observer-based control technique is introduced. Based on the gradient descent method, the control input u_i is formulated as

$$u_i \in -[\nabla f_i(y_i) + \sigma_i(t) \partial G_i(x_i)] - \hat{d}_i(t), \quad (7)$$

where $y_i = \text{col}(y_{i1}, \dots, y_{iN})$, $\hat{d}_i(t)$ is the disturbance observer, and $\partial G_i(x_i)$ is the subgradient of $G_i(x_i)$.

By the disturbance-observer-based control technique, $\hat{d}_i(t), i \in \mathcal{V}$ is generated by

$$\begin{aligned} \dot{\hat{d}}_i(t) &= w_i(t) + M[x_i(t) + v_i(t)], \\ \dot{w}_i(t) &= S w_i(t) + S M[x_i(t) + v_i(t)], \end{aligned} \quad (8)$$

where M is a control gain matrix that satisfies $S - M$ is Hurwitz and $v_i(t)$ is an auxiliary variable generated by

$$\dot{v}_i(t) \in \nabla f_i(y_i) + \sigma_i(t) \partial G_i(x_i). \quad (9)$$

Denote $\bar{d}_i(t) = \hat{d}_i(t) - d_i(t)$ as disturbance observation error. According to the above methods, the circuit implementation of the proposed adaptive distributed approach is shown as Fig. 1. From (2), (3), and (6)-(9), the adaptive distributed approach can be rewritten in the compact form as follows

$$\begin{cases} \dot{x} = -[\nabla f(y) + C \partial G(x)] - \bar{d} \\ \dot{\sigma} = G(x) \\ \dot{\bar{d}} = (I_N \otimes (S - M)) \bar{d} \\ \varepsilon \dot{y} = -H^{\delta(t)} y + B_0^{\delta(t)} (I_N \otimes x), \end{cases} \quad (10)$$

where

$$\begin{aligned} y &= \text{col}(y_1, \dots, y_N), \nabla f(y) = \text{col}(\nabla f_1(y_1), \dots, \nabla f_N(y_N)), \\ \sigma &= \text{col}(\sigma_1, \dots, \sigma_N), G(x) = \text{col}(G_1(x_1), \dots, G_N(x_N)), \\ C &= \text{diag}\{\sigma\} \otimes I_m, \partial G(x) = \text{col}(\partial G_1(x_1), \dots, \partial G_N(x_N)), \\ B_0^{\delta(t)} &= \text{diag}\{a_{11}^{\delta(t)} \otimes I_m, \dots, a_{1N}^{\delta(t)} \otimes I_m, \dots, a_{N1}^{\delta(t)} \otimes I_m, \dots, \\ &a_{NN}^{\delta(t)} \otimes I_m\}, H^{\delta(t)} = L^{\delta(t)} \otimes I_{Nm} + B_0^{\delta(t)}, \\ \bar{d} &= \text{col}(\bar{d}_1, \dots, \bar{d}_N), \sigma^* = \text{col}(\sigma_1^*, \dots, \sigma_N^*). \end{aligned}$$

Since the $\mathcal{G}_p$ ($p \in \mathcal{M}_c$) is directed strongly connected, $-H^p$ is Hurwitz.

Lemma 1. (Khalil, 2002, Theorem 4.6) If a matrix A is Hurwitz, then there exist positive definite symmetric matrices P and Q such that $A^T P + P A = -Q$.

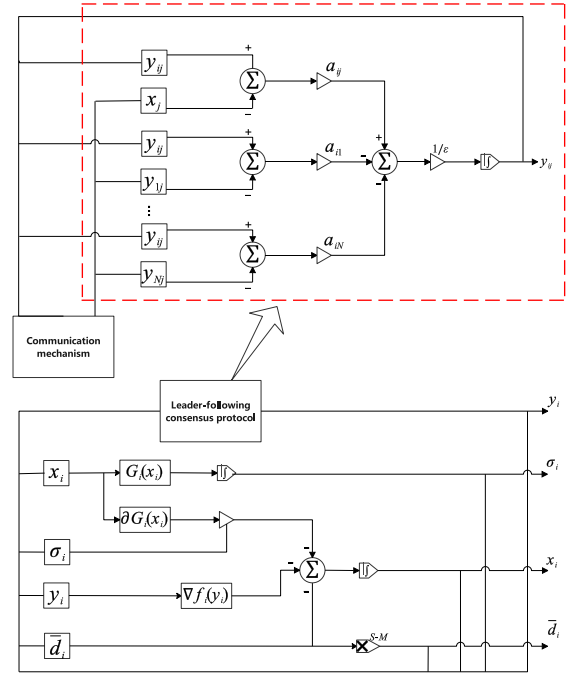


Fig. 1. Circuit implementation of the proposed adaptive distributed NE seeking neurodynamic approach.

Lemma 2. (Gronwall's inequality) If there exist real-valued nonnegative continuous function $\phi(t)$ satisfying

$$\phi(t) \leq u(t) + \int_{t_0}^t \phi(s) p(s) ds \quad \forall t \geq t_0,$$

where $u(t)$ and $p(t)$ are continuous functions, then $\phi(t) \leq u(t) \exp\left(\int_{t_0}^t p(s) ds\right)$.

To clearly describe the feature of switching topologies, the frequency of switching is denoted as

$$F_a(T_1, T_2) = \frac{N_a(T_1, T_2)}{T_2 - T_1},$$

where $N_a(T_1, T_2)$ represents the number of switching during $[T_1, T_2]$.

3.2. Convergence analysis of system

(10) Due to the definition of ε , a two-time-scale structure is introduced. In the original system (10), let $\varepsilon = 0$. There exists at least one real root such that $\mathbf{0} = -H^{\delta(t)} y + B_0^{\delta(t)} (I_N \otimes x)$, which satisfies Assumption 1.1 from Kokotovic (1987). Therefore, for limited systems, the system (10) can be decomposed by applying the singular perturbation technique below.

When $\varepsilon = 0$, the quasi-steady state $y^e = I_N \otimes x$ is obtained for all $\delta(t)$. The reduced system is given as

$$\begin{cases} \dot{x} = -[F(x) + C \partial G(x)] - \bar{d}, \\ \dot{\sigma} = G(x), \\ \dot{\bar{d}} = (I_N \otimes (S - M)) \bar{d} \end{cases} \quad (11)$$

When $\varepsilon \neq 0$, let $\tau = (t - t_0)/\varepsilon$, the boundary-layer system is given below

$$\frac{dy}{d\tau} = -H^{\zeta(\tau)} y + B_0^{\zeta(\tau)} (I_N \otimes x), \quad (12)$$

where x is considered as a constant and $\zeta(\tau) = \delta(\varepsilon\tau + t_0)$ is a piecewise switching signal. According to the transformation of τ , denote τ_p as a

switching time sequence under the fast time-scale. Similar to $\{t_p\}$ shown in Section 2, when $\tau \in [\tau_p, \tau_{p+1})$, $G_{\zeta(\tau)} \in \mathcal{M}_c$ remains unchanged.

Firstly, the stability of the boundary-layer system (12) is analyzed.

Lemma 3. Under Assumptions 1–2, the boundary-layer system (12) is exponentially stable and the estimated information y converges to the actual position $1_N \otimes x$, if there exists $\alpha_1 > 0$, such that for $\forall t$,

$$F_a(t_0, t) \in \ln \rho_1 - \mu_1 \leq -\alpha_1 \quad (13)$$

where

$$\rho_1 = \frac{\max_{\zeta(\tau) \in \mathcal{M}_c} \left\{ \lambda_{\max} \left(P_1^{\zeta(\tau)} \right) \right\}}{\min_{\zeta(\tau) \in \mathcal{M}_c} \left\{ \lambda_{\min} \left(P_1^{\zeta(\tau)} \right) \right\}}$$

and

$$\mu_1 = \frac{\min_{\zeta(\tau) \in \mathcal{M}_c} \left\{ \lambda_{\min} \left(P_1^{\zeta(\tau)} \right) \right\}}{\max_{\zeta(\tau) \in \mathcal{M}_c} \left\{ \lambda_{\min} \left(Q_1^{\zeta(\tau)} \right) \right\}}$$

with positive definite symmetric matrices $P_1^{\zeta(\tau)}$ and $Q_1^{\zeta(\tau)}$ from Hurwitz matrix $-H^{\zeta(\tau)}$.

Proof. Let $\bar{y} = y - 1_N \otimes x$, one obtains

$$\frac{d\bar{y}}{d\tau} = -H^{\zeta(\tau)} \bar{y}.$$

Due to the Hurwitz matrix $-H^{\zeta(\tau)}$, define the multiple Lyapunov function as

$$V_1 = \bar{y}^T P_1^{\zeta(\tau)} \bar{y},$$

where $P_1^{\zeta(\tau)}$ satisfies $-H^{\zeta(\tau)T} P_1^{\zeta(\tau)} - P_1^{\zeta(\tau)} H^{\zeta(\tau)} = -Q_1^{\zeta(\tau)}$. Then one obtains the time derivative of V_1 along (12) below

$$\begin{aligned} \dot{V}_1 &= -2\bar{y}^T P_1^{\zeta(\tau)} H^{\zeta(\tau)} \bar{y} \\ &= \bar{y}^T \left(-H^{\zeta(\tau)T} P_1^{\zeta(\tau)} - P_1^{\zeta(\tau)} H^{\zeta(\tau)} \right) \bar{y} \\ &= -\bar{y}^T Q_1^{\zeta(\tau)} \bar{y} \leq -\lambda_{\min} \left(Q_1^{\zeta(\tau)} \right) \|\bar{y}\|^2 \\ &\leq -\mu_1 V_1, \end{aligned}$$

For each $[\tau_p, \tau_{p+1})$, one obtains $V_1(\tau_{p+1}) \leq e^{-\mu_1(\tau_{p+1}-\tau_p)} V_1(\tau_p)$. Moreover, by the definition of V_1 , there exists ρ_1 such that $V_1(\tau_p) \leq \rho_1 V_1(\tau_p^-)$ for each switching time τ_p . Then

$$\begin{aligned} V_1(\tau) &\leq \rho_1^{N_a(0,\tau)} e^{-\mu_1 \tau} V_1(0) \\ &= e^{N_a(0,\tau) \ln \rho_1} e^{-\mu_1 \tau} V_1(0). \end{aligned} \quad (14)$$

With the transformation of τ , if the condition (13) is satisfied, there exists α_1 such that

$$V_1(t) \leq e^{-\alpha_1(t-t_0)} V_1(t_0), \forall t > t_0,$$

which means the boundary-layer system (12) is exponentially stable and y exponentially converges to $1_N \otimes x$.

□

Then the stability of reduced system (11) is analyzed as follows.

Lemma 4. Under Assumptions 1–3, $(x^*, \sigma^*, \mathbf{0}_{Nm})$ is the equilibrium point of the reduced system (11), if and only if x^* is the NE of the noncooperative game (1).

It is obvious that based on Karush-Kuhn-Tucker conditions, the Lemma holds. We omit the proof due to space limitations.

Lemma 5. Under Assumptions 1–3, the reduced system (11) is stable and $(x, \sigma, \bar{d})$ converges to $(x^*, \sigma^*, \mathbf{0}_{Nm})$, in which x exponentially converges to x^* .

Proof. Let $\bar{x} = x - x^*$, $\bar{\sigma} = \sigma - \sigma^*$, the reduced system (11) can be transformed as follows

$$\begin{cases} \dot{\bar{x}} = -(F(x) - F(x^*) + \Gamma(x)) - \bar{d} \\ \dot{\bar{\sigma}} = G(x) - G(x^*) \\ \dot{\bar{d}} = (I_N \otimes (S - M))\bar{d}, \end{cases} \quad (15)$$

where $\Gamma(x) = C \partial G(x) - C^* \partial G(x^*)$ and $C^* = \text{diag}(\sigma^*) \otimes I_m$.

Due to the Hurwitz matrix $S - M$, there exist positive definite symmetric matrices P_2 and Q_2 satisfying

$$(S - M)^T P_2 + P_2 (S - M) = -Q_2. \quad (16)$$

Define the candidate Lyapunov function as

$$V_2 = \frac{\gamma}{2} \bar{x}^T \bar{x} + \bar{d}^T [I_N \otimes P_2] \bar{d} + \frac{\gamma}{2} \bar{\sigma}^T \bar{\sigma}.$$

Then, the time derivative of V_2 along (15) is given below

$$\begin{aligned} \dot{V}_2 &= -\gamma \bar{x}^T (h - h^* + \bar{\Gamma}) - \bar{x}^T \bar{d} \\ &\quad + \bar{d}^T [I_N \otimes ((S - M)^T P_2 + P_2 (S - M))] \bar{d} \\ &\quad + \gamma (\sigma(t) - \sigma^*) (G(x) - G(x^*)), \end{aligned}$$

where measurable solutions $h \in F(x)$, $h^* \in F(x^*)$ and $\bar{\Gamma} = C\varphi - C\varphi^*$ with $\varphi \in \partial G(x)$, $\varphi^* \in \partial G(x^*)$ satisfy

$$\dot{\bar{x}} = -(h - h^* + \bar{\Gamma}) - \bar{d}.$$

Under Assumption 2, it is easy to obtain

$$-\gamma \bar{x}^T (h - h^*) \leq -\gamma q \|\bar{x}\|^2. \quad (17)$$

By Young's inequality, there exists ϵ_1 such that

$$-\bar{x}^T \bar{d} \leq \|\bar{x}\| \|\bar{d}\| \leq \frac{\epsilon_1}{2} \|\bar{x}\|^2 + \frac{1}{2\epsilon_1} \|\bar{d}\|^2.$$

From (16), one obtains that

$$\begin{aligned} &\bar{d}^T [I_N \otimes ((S - M)^T P_2 + P_2 (S - M))] \bar{d} \\ &= -\bar{d}^T [I_N \otimes Q_2] \bar{d} \leq -\lambda_{\min}(Q_2) \|\bar{d}\|^2. \end{aligned}$$

From the convexity of $G(x)$, one can get

$$\begin{aligned} &-\gamma \bar{x}^T \bar{\Gamma} + \gamma (\sigma(t) - \sigma^*) (G(x) - G(x^*)) \\ &= \gamma \left[\sum_{i=1}^N \bar{x}_i^T (\sigma_i^* \varphi^* - \sigma_i \varphi) + \sum_{i=1}^N (\sigma_i - \sigma_i^*) (G_i(x_i) - G_i(x_i^*)) \right] \\ &\leq \gamma \left[\sum_{i=1}^N (\sigma_i^* (G_i(x_i) - G_i(x_i^*)) - \sigma_i (G_i(x_i) - G_i(x_i^*))) \right. \\ &\quad \left. + \sum_{i=1}^N (\sigma_i - \sigma_i^*) (G_i(x_i) - G_i(x_i^*)) \right] = 0. \end{aligned}$$

Then, one obtains that

$$\dot{V}_2 \leq -\left(\gamma q - \frac{\epsilon_1}{2}\right) \|\bar{x}\|^2 - \left(\lambda_{\min}(Q_2) - \frac{1}{2\epsilon_1}\right) \|\bar{d}\|^2.$$

Choose parameters ϵ_1 and γ such that $c_1 = \gamma q - \frac{\epsilon_1}{2} > 0$ and $c_2 = \lambda_{\min}(Q_2) - \frac{1}{2\epsilon_1} > 0$. Thus, the reduced system (11) is stable and we have $\dot{V}_2 \leq -c_1 \|\bar{x}\|^2$.

Integrating $\dot{V}_2$ with respect to t , one drives that

$$\|\bar{x}\|^2 \leq \frac{2}{\gamma} V_2(t) \leq \frac{2}{\gamma} V_2(t_0) - \mu_2 \int_{t_0}^t \|\bar{x}\|^2 ds,$$

where $\mu_2 = 2c_1/\gamma$. By Lemma 2, we can get $\|\bar{x}\|^2 \leq \frac{2}{\gamma} V_2(t_0) \exp\{-\mu_2 t\}$, which means the player's actions in the reduced system (11) can converge exponentially to NE x^* of game (1). □

By the singular perturbation technique, the original system (10) is analyzed based on Lemmas 3–5.

Theorem 1. Under Assumptions 1–3, if the condition (13) is satisfied, there exists ϵ^* such that when $0 < \epsilon \leq \epsilon^*$, the action profile x from original system (10) exponentially converges to the NE of the noncooperative game (1).

Proof. The stability of the reduced system and the boundary-layer system are proved from Lemmas 3 and 5, respectively. According to Theorem 11.4 in Khalil (2002), the original system (10) is stable and the action profile x exponentially converges to the NE of the noncooperative game (1). $\square$

Remark 2. Theorem 1 demonstrates that, in the absence of causative attacks, the neurodynamic approach achieves Nash equilibrium seeking with disturbance rejection, provided the switching frequency $F_a(t_0, t)$ satisfies condition (13). However, it is important to note that in game-theoretic scenarios where players must continually adjust their actions based on the latest estimates, an excessively high switching frequency may cause the switching process itself to be perceived as continuous disturbances to the update dynamics. Moreover, since not all graphs within the set C_c are necessarily utilized in practice, condition (13) can be considered stringent. To address this concern in practical applications, a milder condition can be employed by selecting parameters ρ_1 and μ_1 that account for scenarios utilizing all possible communication topologies within the set.

4. Adaptive distributed neurodynamic approach with discrete-time switching communication

In this section, the communication scenario under causative attack is introduced, and an adaptive distributed NE seeking neurodynamic approach with discrete-time switching communication is proposed. Besides, the convergence of the proposed approach is analyzed in Section 3, and the Zeno behavior is strictly excluded.

4.1. Adaptive distributed approach with discrete-time communication

Before considering the causative attack, the scenario of the system under a causative attack is introduced. Without loss of generality, the attacker is assumed to be energy-limited. Therefore, generally speaking, the attack is non-sustained. Attacks on certain nodes can lead to communication loss, preventing some nodes from sending or receiving messages. This results in the communication topology switching to a directed non-strongly connected graph, such as a weakly connected or even disconnected graph.

In this situation, the consensus part of the existing distributed algorithms is severely affected (Chen et al., 2024; Deng & Nian, 2018; Gadjev & Pavel, 2019; He & Huang, 2024; Liu & Yi, 2023; Romano & Pavel, 2020; Ye et al., 2023; Ye & Hu, 2017, 2018, 2021; Zhang et al., 2021). After a communication loss occurs, the attacked node immediately attempts to recover communication by restarting the device. The communication recovery time is determined by the initialization time of the communication device and the time required to establish authentication and encryption and is therefore fixed. In this paper, denote the recovery time as t_r .

Based on the above description, there exists at least N directed non-strongly connected graphs after a causative attack for each communication topology $\mathcal{G}_p$, $p \in \mathcal{M}_c$. Denote the set of directed non-strongly connected graphs as C_p , and the index set is represented as $\mathcal{M}_p = \{r+1, \dots, r+N, \dots\}$.

Due to the random retention time of the communication topology, the quantitative relationship between Δ_{t_p} and t_r is not monolithic, and there are three common situations, as shown in Fig. 2. It is considered that one causative attack may affect one or more time intervals of switching topologies.

In order to characterize the causative attack, some relevant features are provided below.

The frequency of causative attacks is denoted as

$$F_b(T_1, T_2) = \frac{N_b(T_1, T_2)}{T_2 - T_1},$$

where $N_b(T_1, T_2)$ represents the total number of causative attacks during $[T_1, T_2]$. And the ratio of time under causative attacks is denoted as $R_p(T_1, T_2) = \frac{T_p(T_1, T_2)}{T_2 - T_1}$, where $T_p(T_1, T_2)$ is the total time under causative attacks.

Remark 3. Fig. 2 shows three situations of causative attack under switching topologies. In situation (a), the causative attack impacts only one switching time interval. In the other two situations, the causative attack impacts several switching time intervals, which expands the causative attacks described in Ye and Hu (2018). Moreover, when the time interval of two causative attacks is less than t_r , the causative attack can be recognized as sustained. It is obvious that $T_p(T_1, T_2) \leq N_b(T_1, T_2) * t_r$.

When causative attacks occur, the communication topology switches between $C_c \cup C_p$, and the switching time sequence is reset to $\{t'_p\}$. Meanwhile, a dynamic event-triggered mechanism is proposed to achieve discrete-time communication, which can effectively reduce the communication cost. Denote the trigger time sequence as $\{t_k^i\}$. Then, the triggered instant t_{k+1}^i is determined by the following condition

$$t_{k+1}^i = \inf\{t > t_k^i \mid \phi \|e_i^y(t)\|^2 > \eta_i(t)\}, \quad (18)$$

where $e_{ij}^y = y_{ij}(t_k^i) - y_{ij}(t)$, $e_i^y = y_i(t_k^i) - y_i(t)$, the constant parameter ϕ is defined later and the internal dynamical variable η_i satisfies that

$$\dot{\eta}_i(t) = -g_i \eta_i(t) - \phi \|e_i^y(t)\|^2, \quad (19)$$

with $g_i > 0$ being a constant and $\eta_i(t_0) > 0$.

Note that both event-triggered sequence $\{t_k\}$ and switching sequence $\{t'_p\}$ are considered, and the information broadcasting occurs only at triggered instants in this paper. When the communication topologies change between two consecutive event-triggered instants, the actual switching topologies may differ from the preset ones, which is shown in Example 1 in Ding et al. (2024b). To satisfy the need for players to sample information at switching instants, the switching-based compensation communication mechanism is introduced from Ding et al. (2024b). This mechanism requires players to retransmit the last sampled information from the previous trigger to the new neighbors (including those affected by the switching of communication topology and communication recovery).

Thus, the adaptive distributed approach with discrete-time switching communication under causative attack is proposed as follows

$$\begin{cases} \dot{x}_i \in [\nabla f_i(y_i) + \sigma_i \partial G_i(x_i)] - \bar{d}_i \\ \dot{\sigma}_i = G_i(x_i) \\ \dot{\bar{d}}_i = (S - M)\bar{d}_i \\ \varepsilon \dot{y}_{ij}(t) = - \left(\sum_{r=1}^N a_{ir}^{\Delta(t)} (y_{ij}(t_k^i) - y_{rj}(t_k^i)) - a_{ij}^{\Delta(t)} (y_{ij}(t_k^i) - x_j(t_k^i)) \right), \end{cases} \quad (20)$$

where $\Delta(t)$ satisfies

$$\begin{cases} \Delta(t) = \delta(t), & \text{if no causative attack,} \\ \Delta(t) \in \mathcal{M}_p, & \text{otherwise.} \end{cases}$$

Define $e_i^x = x_i(t_k^i) - x_i(t)$, $e^x = \text{col}(e_1^x, \dots, e_N^x)$ and $e^y = \text{col}(e_1^y, \dots, e_N^y)$. From (17), the compact form of the neurodynamic approach (20) is given

$$\begin{cases} \dot{x} = -(h - h^* + \bar{\Gamma}) - \bar{d} \\ \dot{\sigma} = G(x) \\ \dot{\bar{d}} = (I_N \otimes (S - M))\bar{d} \\ \varepsilon \dot{y} = -H^{\Delta(t)}(y + e^y) + B_0^{\Delta(t)}(I_N \otimes (x + e^x)), \end{cases} \quad (21)$$

where $B_0^{\Delta(t)} = \text{diag}\{a_{11}^{\Delta(t)} \otimes I_m, \dots, a_{1N}^{\Delta(t)} \otimes I_m, \dots, a_{N1}^{\Delta(t)} \otimes I_m, \dots, a_{NN}^{\Delta(t)} \otimes I_m\}$, $H^{\Delta(t)} = L^{\Delta(t)} \otimes I_{Nm} + B_0^{\Delta(t)}$. When $\Delta(t) \in \mathcal{M}_p$, $-H^{\Delta(t)}$ is not Hurwitz.

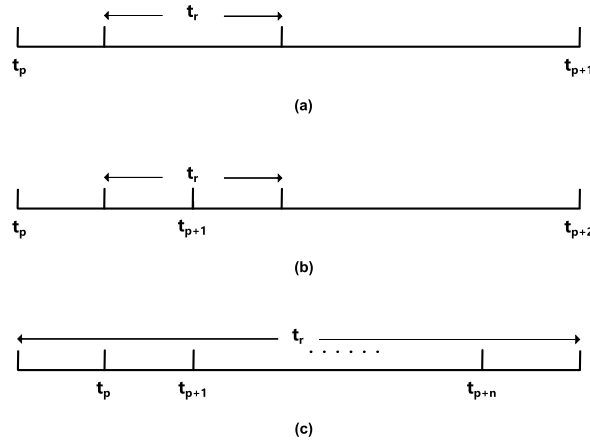


Fig. 2. Three situations of causative attack under switching topologies: A causative attack occurs: (a) in a time interval of switching (t_p, t_{p+1}) , (b) before a switching time t_{p+1} , and the time interval between the attacked time and t_{p+1} is less than t_r , (c) before a switching time t_p , and there exists $n \geq 2$ such that $t_r \geq t_{p+n} - t_p$.

Remark 4. In (20), a good action estimation approach is utilized. Compared to the effective action estimation approach from Ye and Hu (2017), the proposed approach incorporates a dynamic event-triggering mechanism (18) and (19). This not only avoids continuous communication but further reduces the frequency of discrete communications through dynamic threshold adjustment. In addition, the introduction of singular perturbation parameter ε constructs a two-time-scale structure, which enhances the action estimation approach by accelerating the consensus speed.

4.2. Convergence analysis of system

(21) Similar to the system (10), the original system (21) is divided into two systems. The reduced system remains unchanged as (11), and the boundary-layer system is given below

$$\frac{dy}{d\tau} = -H^{\Phi(\tau)}(e^y + y) + B_0^{\Phi(\tau)}(1_N \otimes (x + e^x)), \quad (22)$$

where $\Phi(\tau) = \Delta(\varepsilon\tau + t_0)$ is a piecewise switching signal. With the fast time-scale, the new time sequences are denoted as $\{\tau'_p\}$ and $\{\tau'_k\}$ derived from $\{t'_p\}$ and $\{t'_k\}$.

Theorem 2. Under Assumptions 1–3, there exists ε^* such that when $0 < \varepsilon \leq \varepsilon^*$, the action profile x from the original system (21) with the event-triggered mechanism (18) exponentially converges to the NE of the noncooperative game (1), if there exists $\alpha_2 > 0$, such that

$$F_a(t_0, t) \varepsilon \ln \rho_2 + F_b(t_0, t) \varepsilon [\ln \rho_2 + (\mu_3 + \theta)t_r] - \mu_3 \leq -\alpha_2, \quad (23)$$

where

$$\rho_2 = \frac{\max \left\{ 1, \max_{\Phi(\tau) \in \mathcal{M}_c} \left\{ \lambda_{\max} \left(P_1^{\Phi(\tau)} \right) \right\} \right\}}{\min \left\{ 1, \min_{\Phi(\tau) \in \mathcal{M}_c} \left\{ \lambda_{\min} \left(P_1^{\Phi(\tau)} \right) \right\} \right\}},$$

$$\mu_3 = \frac{\min_{\Phi(\tau)} \left\{ c_3^{\Phi(\tau)}, \min_i \{g_i\} \right\}}{\max_{\Phi(\tau)} \left\{ \lambda_{\min} \left(P_1^{\Phi(\tau)} \right), 1 \right\}}$$

with $c_3^{\Phi(\tau)} = \lambda_{\min} \left(Q_1^{\Phi(\tau)} \right) - \frac{2 \| P_1^{\Phi(\tau)} H^{\Phi(\tau)} \|^2}{\varepsilon_2}$ and $\theta = \max_{\Phi(\tau) \in \mathcal{M}_p} \{c_4^{\Phi(\tau)}\}$ with

$$c_4^{\Phi(\tau)} = 2 \| H^{\Phi(\tau)} \|^2 + \frac{2 \| H^{\Phi(\tau)} \|^2}{\varepsilon_3}.$$

Proof.

Let $\bar{y} = y - 1_N \otimes (x + e^x)$, the system (22) can be transformed as

$$\frac{d\bar{y}}{d\tau} = -H^{\Phi(\tau)}(e^{\bar{y}} + \bar{y}).$$

Due to the triggered condition (18), the dynamics of η_i defined in (19) have

$$\dot{\eta}_i(\tau) = -g_i \eta_i(\tau) - \phi \| e_i^y(\tau) \|^2 \geq -(1 + g_i) \eta_i(\tau).$$

Thus, $\eta_i(\tau) \geq e^{-(1+g_i)\tau} \eta_i(0) > 0$.

Depending on whether the causative attacks occur, the boundary-layer system is discussed in two different cases. Define the multiple Lyapunov function as follows

$$V_3 = \begin{cases} \bar{y}^T P_1^{\Phi(\tau)} \bar{y} + \sum_{i=1}^N \eta_i, & \Phi(\tau) \in \mathcal{M}_c, \\ \bar{y}^T \bar{y} + \sum_{i=1}^N \eta_i, & \Phi(\tau) \in \mathcal{M}_p. \end{cases}$$

When $\Phi(\tau) \in \mathcal{M}_c$, one can deduce from (19) and Young's inequality that there exists ε_2 such that

$$\begin{aligned} \dot{V}_3 &= 2\bar{y}^T P_1^{\Phi(\tau)} (-H^{\Phi(\tau)}(e^{\bar{y}} + \bar{y})) + \sum_{i=1}^N \dot{\eta}_i \\ &\leq - \left(\lambda_{\min} \left(Q_1^{\Phi(\tau)} \right) - \frac{2 \| P_1^{\Phi(\tau)} H^{\Phi(\tau)} \|^2}{\varepsilon_2} \right) \|\bar{y}\|^2 \\ &\quad + \frac{\varepsilon_2}{2} \|e^{\bar{y}}\|^2 - \sum_{i=1}^N g_i \eta_i - \phi \|e^{\bar{y}}\|^2. \end{aligned}$$

Let $\phi = \frac{\varepsilon_2}{2}$, one obtains

$$\dot{V}_3 \leq -c_3^{\Phi(\tau)} \|\bar{y}\|^2 - \sum_{i=1}^N g_i \eta_i.$$

Choose parameter ε_2 satisfies $c_3^{\Phi(\tau)} > 0$, $\Phi(\tau) \in \mathcal{M}_c$, thus $\dot{V}_3 \leq -\mu_3 V_3$. For $\tau \in [\tau'_p, \tau'_{p+1})$, $\mathcal{G}_{\Phi(\tau)}$ remains unchanged and $V_3(\tau) \leq e^{-\mu_3(\tau - \tau'_p)} V_3(\tau'_p)$.

When $\Phi(\tau) \in \mathcal{M}_p$, $\dot{V}_3 = 2\bar{y}^T (-H^{\Phi(\tau)}(e^{\bar{y}} + \bar{y})) + \sum_{i=1}^N \dot{\eta}_i$. By Young's inequality,

$$\begin{aligned} -2\bar{y}^T H^{\Phi(\tau)} e^{\bar{y}} &\leq 2 \| H^{\Phi(\tau)} \| \|\bar{y}\| \|e^{\bar{y}}\| \\ &\leq \frac{2 \| H^{\Phi(\tau)} \|^2}{\varepsilon_3} \|\bar{y}\|^2 + \frac{\varepsilon_3}{2} \|e^{\bar{y}}\|^2. \end{aligned}$$

From (19), one can get that

$$\dot{V}_3 \leq c_4^{\Phi(\tau)} \|\bar{y}\|^2 - \sum_{i=1}^N g_i \eta_i.$$

Thus, we have $\dot{V}_3 \leq \theta V_3$, which implies that $V_3(\tau) \leq e^{\theta(\tau - \tau'_p)} V_3(\tau'_p)$ for $\forall \tau \in [\tau'_p, \tau'_{p+1})$.

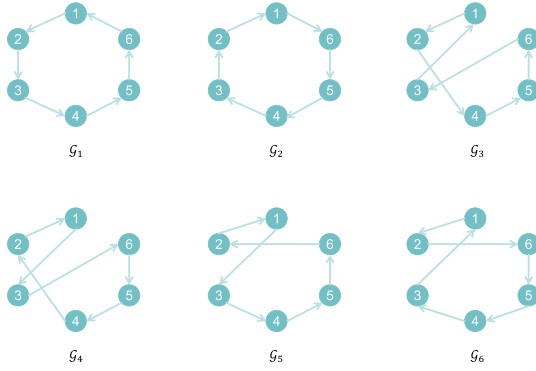


Fig. 3. A set of six directed strongly connected graphs.

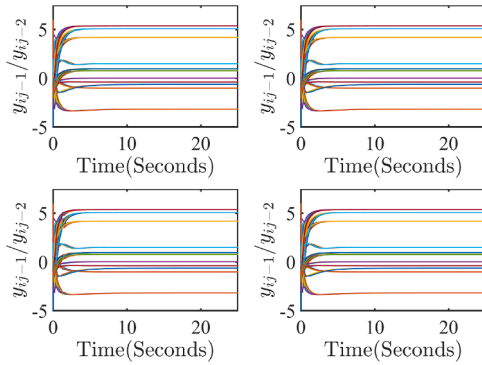


Fig. 4. Convergences of six vehicles actions' estimation $y_{ij} = [y_{ij-1}, y_{ij-2}]^T \in \mathbb{R}^2, i, j \in 1, 2, \dots, 6$ in Case 1.

Consider the time interval $[0, \tau]$. When $\Phi(\tau) \in \mathcal{M}_c$, we have $V_3(\tau'_{p+1}) \leq e^{-\mu_3(\tau'_{p+1} - \tau'_p)} V_3(\tau'_p)$. And when $\Phi(\tau) \in \mathcal{M}_p$, we have $V_3(\tau'_{p+1}) \leq e^{\theta(\tau'_{p+1} - \tau'_p)} V_3(\tau'_p)$. Moreover, by the definition of V_3 , there exists ρ_2 such that $V_3(\tau'_p) \leq \rho_2 V_3(\tau'_p)$ for each switching time τ'_p . For $\forall \tau > 0$, we have

$$V_3(\tau) \leq \rho_2^{N_a(0, \tau) + N_b(0, \tau)} e^{-\mu_3(\tau - T_p(0, \tau))} e^{\theta T_p(0, \tau)} V_3(0) \\ = e^{(N_a(0, \tau) + N_b(0, \tau)) \ln \rho_2 - \mu_3(\tau - T_p(0, \tau)) + \theta T_p(0, \tau)} V_3(0).$$

From the condition (23) and the transformation of τ , one obtains

$$V_3(t) \leq e^{-\alpha_2(t-t_0)} V_3(t_0), \forall t > t_0.$$

which means the boundary-layer system (22) is exponentially stable and $\bar{y}$ exponentially converges to $\mathbf{0}_{N^2m}$.

From Lemma 4, the reduced system is stable. Similar to Theorem 1, the original system (21) with the event-triggered mechanism (18) is stable. And the action profile x exponentially converges to the NE of the noncooperative game (1).

□

Remark 5. Theorem 2 shows that, for a given switching frequency, when the frequency of causative attacks $F_b(t_0, t)$ and the communication recovery time t_r satisfy condition (23), the original system is resilient to causative attacks. Clearly, when the resilience of the communication device improves (i.e., t_r is shorter), the system can withstand a higher frequency of causative attacks. Moreover, in practical applications, the parameters in condition (23) can be chosen as mild as those in Remark 2.

In the event-triggered process, Zeno behavior, an extreme situation where a player infinitely triggers within a finite time interval, can severely impact the stability of the closed-loop system. To ensure practical applicability, the Zeno behavior must be strictly excluded. Next, we

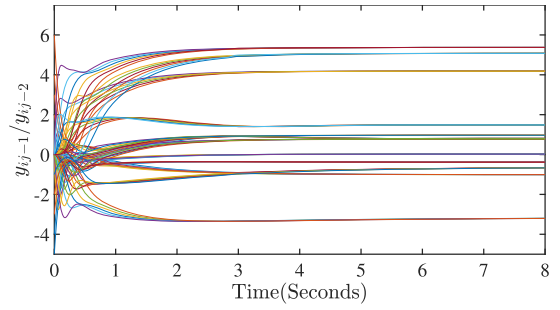


Fig. 5. The details of the first subfigure in Fig. 4, showing the fluctuations caused by topology switching.

will analyze the proposed event-triggered mechanism and demonstrate that it is Zeno-free.

Lemma 6. Under the condition of Theorem 2, the proposed event-triggered mechanism (18) will not occur Zeno behavior.

Proof. Without loss of generality, we demonstrate the exclusion of Zeno behavior under the normal time-scale. By contradiction, suppose that Zeno behavior occurs. Then, there exists a positive constant T' such that $\lim_{k \rightarrow \infty} t_k^i = T'$.

Let $\varepsilon_0 = \frac{\sqrt{\eta_i(0)}}{2M_0\sqrt{\phi}} e^{-\frac{1}{2}(1+g_i)T'} > 0$, there must exist $k_0 > 0$ such that $t_k^i \in [T' - \varepsilon_0, T']$, $\forall k \geq k_0$. For $t \in (t_{k_0}^i, t_{k_0+1}^i)$, we have

$$\phi \|y_i(t_k^i) - y_i(t)\|^2 \leq \eta_i(t)$$

from the event-triggered condition (18). Obviously, there has $T_0 \in (t_{k_0}^i, t_{k_0+1}^i)$ such that

$$\phi \|y_i(t_{k_0}^i) - y_i(T_0)\|^2 = \eta_i(T_0).$$

According Theorem 2, there exists M_0 such that $\|y_i(t)\| \leq M_0$. Thus,

$$\phi \|y_i(t_{k_0}^i) - y_i(T_0)\|^2 \\ \leq \phi \left\| \int_{t_{k_0}^i}^{T_0} \dot{y}_i(t) dt \right\|^2 \leq \phi M_0^2 (T_0 - t_{k_0}^i)^2.$$

Note that

$$\eta_i(T_0) \geq e^{-(1+g_i)T_0} \eta_i(0) \geq e^{-(1+g_i)T'} \eta_i(0) = 4M_0^2 \phi \varepsilon_0^2.$$

Then, one obtains

$$t_{k_0+1}^i - t_{k_0}^i > T_0 - t_{k_0}^i \geq 2\varepsilon_0,$$

which conflicts with the assumption that $t_{k_0+1}^i \in [T' - \varepsilon_0, T']$. Hence, the event-triggered mechanism (18) is Zeno-free.

□

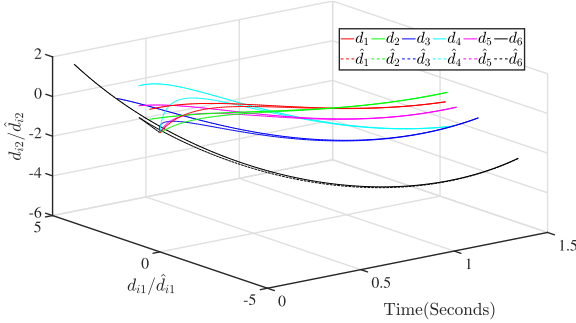
5. Simulation and numerical example

In this section, consider an autonomous vehicle problem (Stankovic et al., 2012) to verify the validity of the designed neurodynamic approaches (10) and (21). In this scenario, multiple vehicles must coordinate their strategies to navigate without collisions while adhering to individual constraints and responding to potential communication disruptions.

A group of six self-driving vehicles connect each other with mobile sensor networks in the noncooperative game (1). And the position of player i is described as $x_i \in \mathbb{R}^2$, i.e., $m = 2$, $x_i = [x_{i1}, x_{i2}]^T, i = 1, 2, \dots, 6$. For each vehicle i , the objective function is formulated as $f_i(x) = \alpha_i x_i^T x_i + \beta_i 1_2^T x_i + \frac{1}{2} \sum_{j=1}^6 m_{ij} \|x_i - x_j\|^2$, where $\alpha_i = \frac{1}{2} * i$ and $\beta_i =$

Table 1The switching process of communication topologies ($\{t_p\}$ denotes the switching sequence for each simulation).

Model	Simulation 1		Simulation 2		Simulation 3		Simulation 4	
	t_p	graph	t_p	graph	t_p	graph	t_p	graph
Case 1: Four simulations of approach (10) ($F_a(0, 25) \leq 0.48$)	2.8535	$G_1 \rightarrow G_4$	1.9123	$G_1 \rightarrow G_3$	1.1306	$G_1 \rightarrow G_6$	1.7339	G_1
	4.5802	G_4	4.2761	$G_3 \rightarrow G_6$	3.7343	$G_6 \rightarrow G_5$	4.8913	G_1
	7.7277	G_4	6.4123	$G_6 \rightarrow G_5$	5.9744	$G_5 \rightarrow G_3$	6.9192	$G_1 \rightarrow G_3$
	9.8948	$G_4 \rightarrow G_5$	7.8802	$G_5 \rightarrow G_6$	8.6540	$G_3 \rightarrow G_4$	10.8698	$G_3 \rightarrow G_4$
	13.2313	$G_5 \rightarrow G_2$	9.3219	$G_6 \rightarrow G_2$	11.1554	$G_4 \rightarrow G_1$	13.7839	$G_4 \rightarrow G_3$
	17.0816	$G_2 \rightarrow G_1$	11.3385	$G_2 \rightarrow G_6$	14.6815	$G_1 \rightarrow G_5$	17.7334	$G_3 \rightarrow G_1$
	19.0725	$G_1 \rightarrow G_2$	13.0655	$G_6 \rightarrow G_5$	15.8394	$G_5 \rightarrow G_6$	19.5559	$G_1 \rightarrow G_5$
	22.2148	$G_2 \rightarrow G_3$	14.4352	$G_5 \rightarrow G_4$	19.0934	G_6	21.9314	$G_5 \rightarrow G_6$
	25.9538		16.5406	$G_4 \rightarrow G_6$	20.2214	$G_6 \rightarrow G_4$	25.8931	
			18.6776	$G_6 \rightarrow G_4$	23.0080	$G_4 \rightarrow G_1$		
			20.9529	$G_4 \rightarrow G_1$	23.2879	$G_1 \rightarrow G_6$		
			22.7549	$G_1 \rightarrow G_3$	26.2339			
			26.2462					
Case 2: Four simulations of approach (21) with event-triggered mechanism (18) and (19) ($F_b(0, 25) = 40$, $F_a(0, 25) \leq 0.48$, $t_r = 2s$)	2.6211	$G_1 \rightarrow G_4$	2.0741	G_1	1.3439	G_1	1.1830	$G_1 \rightarrow G_6$
	3.7983	$G_4 \rightarrow G_3$	5.2501	G_1	3.0370	$G_1 \rightarrow G_5$	3.9563	$G_6 \rightarrow G_1$
	5.2510	$G_3 \rightarrow G_5$	6.6984	G_1	4.1001	$G_5 \rightarrow G_1$	5.9292	$G_1 \rightarrow G_3$
	6.6227	$G_5 \rightarrow G_1$	8.3749	$G_1 \rightarrow G_5$	7.2506	$G_1 \rightarrow G_2$	7.1082	G_3
	9.1798	$G_1 \rightarrow G_6$	10.3107	$G_5 \rightarrow G_2$	9.1006	$G_2 \rightarrow G_3$	8.8205	$G_3 \rightarrow G_2$
	12.5183	$G_6 \rightarrow G_1$	12.2561	$G_2 \rightarrow G_1$	12.8880	$G_3 \rightarrow G_4$	10.4948	$G_2 \rightarrow G_4$
	15.4385	$G_1 \rightarrow G_4$	15.5870	$G_1 \rightarrow G_4$	15.8809	G_4	12.4713	$G_4 \rightarrow G_3$
	18.8628	$G_4 \rightarrow G_2$	18.3472	$G_4 \rightarrow G_5$	19.0267	$G_4 \rightarrow G_1$	13.8944	$G_3 \rightarrow G_1$
	20.0548	$G_2 \rightarrow G_5$	21.3829	$G_5 \rightarrow G_2$	20.5103	$G_1 \rightarrow G_4$	16.6026	$G_1 \rightarrow G_4$
	21.5823	G_5	24.2746	$G_2 \rightarrow G_1$	24.3638	$G_4 \rightarrow G_3$	18.0898	$G_4 \rightarrow G_5$
	25.0108		27.1979		27.5458		20.2529	$G_5 \rightarrow G_1$
							22.2235	$G_1 \rightarrow G_5$
							25.1545	

**Fig. 6.** Trajectories of disturbances $d_i = [d_{i1}, d_{i2}]^T \in \mathbb{R}^2$ and disturbance observers $\hat{d}_i = [\hat{d}_{i1}, \hat{d}_{i2}]^T \in \mathbb{R}^2$ which are depicted by solid and dashed lines, respectively.

$2 * (-1)^i * i^2$ are constants, and according to the willingness of connections between vehicles, m_{ij} satisfies $m_{12} = m_{23} = m_{34} = m_{45} = m_{56} = m_{61} = 1$, otherwise, $m_{ij} = 0$ for $i, j = 1, 2, \dots, 6$. Moreover, the inequality constraints are given below

$$\begin{aligned}
 g_1(x_1) &= \|x_1 - [2, 3]^T\|^2 - 9 \leq 0, \\
 g_2(x_2) &= \|x_2 - [2, 3]^T\|^2 - 16 \leq 0, \\
 g_3(x_3) &= \|x_3 - [-3, 4]^T\|^2 - 4 \leq 0, \\
 g_4(x_4) &= \|x_4 - [2, 3]^T\|^2 - 9 \leq 0, \\
 g_5(x_5) &= \|x_5 - [2, 3]^T\|^2 - 25 \leq 0, \\
 g_6(x_6) &= \|x_6 - [1, -2]^T\|^2 - 4 \leq 0.
 \end{aligned}$$

Solving the autonomous vehicle problem for each vehicle i is equivalent to minimizing their local objective function f_i . According to the above setting, it could be computed that NE of the game (1) is $x^* = [0.81, 1.48, -0.38, 0.97, -1.01, 4.18, 0.02, 0.77, 5.09, 5.38, -0.62, -3.17]^T$. It is presumed that mobile vehicles can exchange information with their adjacent ones through a set of communication topologies depicted in

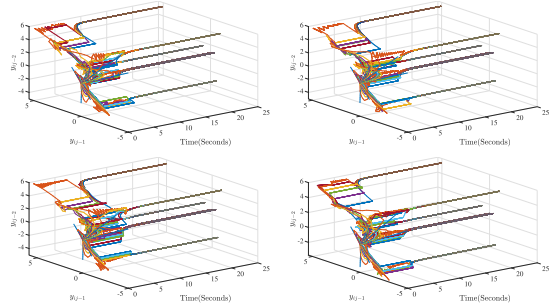
**Fig. 7.** Convergences of six vehicles actions' estimation $y_{ij} = [y_{ij-1}, y_{ij-2}]^T \in \mathbb{R}^2$, $i, j = 1, 2, \dots, 6$ in Case 2.

Fig. 3. Different from Chen et al. (2024), He and Huang (2024), Ding et al. (2024b), Ye and Hu (2021) and Ye and Hu (2018), it is considered that the sequence and the retention time of switching are both random.

Similar to Chen et al. (2024), set $\hat{d}(0) = \mathbf{0}_{12}$, $d(0) = [0.5, 1, 0.5, 0.5, 2, 1, 1, 2, 1, 1, 4, 2]$, and $S = \begin{bmatrix} 0 & 2 \\ -2 & 0 \end{bmatrix}$ in (3) and (8),

and choose $M = \begin{bmatrix} 4 & 2 \\ -2 & 4 \end{bmatrix}$ such that $S - M$ is Hurwitz. The singular perturbation parameter $\varepsilon = 0.1$ is settled. The initials of players are $x(0) = [-1, 3, 1, 2, -5, 4, 0, -3, -4, 6, -1, -2]^T$.

Case 1. Considering that vehicles communicate with each other without causative attacks in a continuous-time manner, we repeat the simulation four times due to the randomness of the switching of communication topologies and set $F_a(0, 25) \leq 0.48$, i.e., the number of switching is not more than 12. Under the proposed neurodynamic approach (10), the simulation results can be illustrated by Figs. 4–6. For the four simulations, the transformations of their communication topology switching are given as part of Case 1 in Table 1.

According to the simulations, all actions (positions) from each simulation converge to $x^* = [0.81, 1.48, -0.38, 0.97, -1.01, 4.18, 0.02, 0.77, 5.09,$

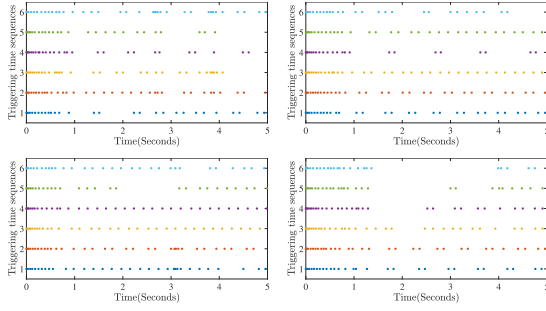


Fig. 8. Triggering time sequences of the six vehicles in Case 2.

$5.38, -0.62, -3.17]^T$ and each player's estimation of another player's action reaches consensus and converges to the actual action. Fig. 4 shows that the switching communication topology primarily affects the update phase of the estimation consistency process but does not impede the final convergence to NE of the game (1). Meanwhile, the estimations on the actions of other players reach the real actions. Fig. 5 zooms in on some of the details of Subfigure 1 of Fig. 4, showing that changes in communication topology can cause fluctuations. Trajectories of external disturbances and disturbance observers are depicted in Fig. 6, in which the disturbance observers (solid lines) effectively track the actual disturbances (dashed lines), demonstrating the successful attenuation of non-vanishing external disturbances.

Case 2. Considering that vehicles communicate with each other under causative attacks in a discrete-time manner, we repeat the simulation four times, as in Case 1. Under the proposed neurodynamic approach (21) with the event-triggered mechanism (18) and (19), the simulation results can be illustrated by Fig. 7–8. In this case, set the communication recovery time $t_r = 2s$ according to the communication device initialization time and the authentication and encryption establishment time (Sousa et al., 2021) and assume that the frequency of causative attacks $F_b(0, 25) = 40$. The transformations of their communication topology switching are given as part of Case 2 in Table 1. Set the parameter $g_i = 0$, for $i \in \mathcal{V}$, $\phi = 0.5$, and the initial value $\eta(0) = \mathbf{1}_6$.

Despite the presence of causative attacks on system (21), the simulation results confirm that all vehicles' actions successfully converge to the same NE as in Case 1. As shown in Fig. 7, the estimations of other players' actions also reach consensus with the true values. Furthermore, Fig. 8 displays the event-triggered instants for each vehicle during the $0 \leq t \leq 5$ interval, verifying the effectiveness of the proposed event-triggered communication mechanism. These collective results demonstrate the robustness and adaptability of the proposed approaches in maintaining stable convergence to the NE even under adversarial communication conditions. Consequently, this case study underscores the practical applicability of the developed methods and reinforces their potential for real-world noncooperative game scenarios with inherent communication vulnerabilities.

Comparison : Under the switching process of communication topologies as Simulation 1 of Case 1 in Table 1, the proposed neurodynamic approach (21) is demonstrated its ability to defend against causative attacks, as shown in Fig. 9. Meanwhile, we compared the approach proposed in this paper with the approach proposed in the literature (Chen et al., 2024) under the condition that the vehicles communicate continuously under the switching process of simulation 1 of Case 1 and without private inequality constraints. Without the private inequality constraints, the actions (positions) converge to $x^* = [0.21, 0.21, -1.59, -1.59, 3.23, 3.23, -5.08, -5.08, 6.62, 6.62, -10.26, -10.26]^T$. The comparison results are displayed in Fig. 10. With the approach proposed in this paper, the vehicles converge faster to reach NE.

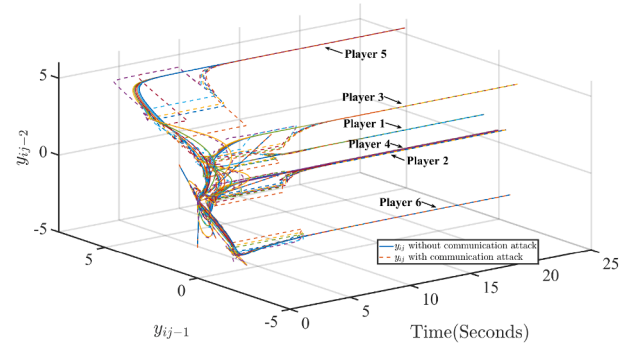


Fig. 9. Comparison between neurodynamic approaches (10) and (21) under the switching situation of communication topologies in Simulation 1 of Case 1.

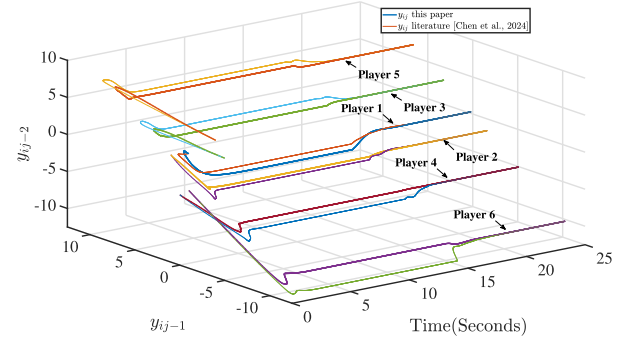


Fig. 10. Comparison between the proposed approach and the literature (Chen et al., 2024) under the switching situation of communication topologies in Simulation 1 of Case 1.

Remark 6. In Fig. 7, a 'zigzag' pattern can be observed in the trajectories. Unlike the discrete update steps of discrete-time algorithms, this 'zigzag' phenomenon is a direct result of the event-triggered mechanism. In this simulation, Algorithm (21) operates under a discrete-time, event-triggered communication mechanism governed by conditions (18) and (19). The discrete, aperiodic jumps in the communicated information cause corresponding abrupt changes in the value of $\dot{y}$ within the algorithm dynamics (21). It is these discrete updates that manifest as the visible 'zigzag' pattern in the trajectory.

Remark 7. In comparison to the switching mechanisms presented in Chai et al. (2024), Chen et al. (2024), Ding et al. (2024b) and Ye and Hu (2018), the proposed mechanism in this paper effectively reduces the risk of information leakage, as the random switching times are significantly less susceptible to persistent information theft by exploratory attackers. This random feature is exemplified in the switching sequences of different simulations of the same case, which are given in Table 1.

6. Conclusions

This paper has presented a novel attack-resilient adaptive distributed neurodynamic approach for solving noncooperative games under adversarial communication environments and local constraints. The main contributions are threefold. First, we proposed an adaptive distributed neurodynamic approach with continuous-time switching communication to achieve NE seeking while ensuring privacy protection against exploratory attacks and rejecting external disturbances. The convergence under random switching topologies was rigorously proved using singular perturbation techniques. Second, to address more challenging scenarios involving causative attacks and discrete communication, we developed an attack-resilient adaptive distributed approach with discrete-time switching communication and an event-triggered mechanism. This approach guarantees exponential convergence to the NE even under

communication attacks and strictly excludes Zeno behavior. Third, the theoretical findings were validated through comprehensive simulations in a self-organizing autonomous vehicle network. The results demonstrated the approach's superior convergence rate, resilience against both exploratory and causative attacks, and effective handling of private inequality constraints compared to existing methods.

CRedit authorship contribution statement

Zhijie Chen: Writing – review & editing, Writing – original draft, Validation, Software, Methodology, Investigation, Formal analysis, Conceptualization; **Jianing Chen:** Writing – review & editing, Methodology, Conceptualization; **Xinwen Bu:** Writing – original draft, Software; **Sitian Qin:** Writing – review & editing, Validation, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

- Bahrami, S., & Parniani, M. (2014). Game theoretic based charging strategy for plug-in hybrid electric vehicles. *IEEE Transactions on Smart Grid*, 5(5), 2368–2375. <https://doi.org/10.1109/TSG.2014.2317523>
- Cai, X., Xiao, F., Wei, B., Yu, M., & Fang, F. (2023). Nash equilibrium seeking for general linear systems with disturbance rejection. *IEEE Transactions on Cybernetics*, 53(8), 5240–5249. <https://doi.org/10.1109/TCYB.2022.3195361>
- Cao, Y., Tang, S., Li, C., Zhang, P., Tan, Y., Zhang, Z., & Li, J. (2012). An optimized EV charging model considering TOU price and SOC curve. *IEEE Transactions on Smart Grid*, 3(1), 388–393. <https://doi.org/10.1109/TSG.2011.2159630>
- Chai, Y., Hu, Y., Qin, S., Feng, J., & Xu, C. (2024). Distributed adaptive event-triggered algorithms for nonsmooth resource allocation optimization over switching topologies. *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, 54(6), 3484–3496. <https://doi.org/10.1109/TSMC.2024.3358405>
- Chen, S., Wan, Y., & Cao, J. (2025). Adaptive Nash equilibrium seeking in hybrid heterogeneous open multi-agent systems under DoS attacks. *IEEE Transactions on Network Science and Engineering*, 12(4), 2770–2782. <https://doi.org/10.1109/TNSE.2025.3553900>
- Chen, W.-H., Yang, J., Guo, L., & Li, S. (2016). Disturbance-observer-based control and related methods: An overview. *IEEE Transactions on Industrial Electronics*, 63(2), 1083–1095. <https://doi.org/10.1109/TIE.2015.2478397>
- Chen, X., Wang, P., & Yu, W. (2024). Distributed NE seeking with disturbance rejection for games in networks under directed switching topology. *IEEE Transactions on Circuits and Systems II: Express Briefs*, 71(7), 3438–3442. <https://doi.org/10.1109/TCSII.2024.3362594>
- Deng, Z., & Nian, X. (2018). Distributed algorithm design for aggregative games of disturbed multiagent systems over weight-balanced digraphs. *International Journal of Robust and Nonlinear Control*, 28, 5344–5357. <https://doi.org/10.1002/rnc.4316>
- Ding, L., Chen, C., Ye, M., & Han, Q.-L. (2024a). Fully distributed Nash equilibrium seeking: A double-layer adaptive approach. *IEEE Transactions on Neural Networks and Learning Systems*, (pp. 1–15). <https://doi.org/10.1109/TNNLS.2024.3489356>
- Ding, L., & Yan, G. (2020). A survey of the security issues and defense mechanisms of multi-agent systems. *CAAI Transactions on Intelligent Systems*, 15(3), 425–434. <https://doi.org/10.11992/tis.201812015>
- Ding, L., Ye, M., & Yue, D. (2024b). Communication-efficient distributed Nash equilibrium seeking under switching topologies: A decentralized gradient-based event-triggered scheme. *Automatica*, 167, 111738. <https://doi.org/10.1016/j.automatica.2024.111738>
- Facchinei, F., & Pang, J.-S. (2009). Nash equilibria: The variational approach. In D. P. Palomar, & Y. C. E. Eldar (Eds.), *Convex optimization in signal processing and communications*, pp. 443–493. Cambridge University Press. <https://doi.org/10.1017/CBO9780511804458.013>
- Feng, S., Ye, M., Xie, L., & Xu, S. (2025). Quantized distributed Nash equilibrium seeking under DoS attacks. *IEEE Transactions on Control of Network Systems*, 12(3), 1856–1868. <https://doi.org/10.1109/TCNS.2025.3543666>
- Gadjov, D., & Pavel, L. (2019). A passivity-based approach to Nash equilibrium seeking over networks. *IEEE Transactions on Automatic Control*, 64(3), 1077–1092. <https://doi.org/10.1109/TAC.2018.2833140>
- Godsil, C., & Royle, G. F. (2001). Algebraic graph theory. New York: Springer Science & Business Media. <https://doi.org/10.1007/978-1-4613-0163-9>
- He, X., & Huang, J. (2024). Distributed Nash equilibrium seeking with dynamics subject to disturbance of unknown frequencies over jointly strongly connected switching networks. *IEEE Transactions on Automatic Control*, 69(1), 606–613. <https://doi.org/10.1109/TAC.2023.3271791>
- He, X., Yu, J., Huang, T., Li, C., & Li, C. (2014). Neural network for solving Nash equilibrium problem in application of multiuser power control. *Neural Networks*, 57, 73–78. <https://doi.org/10.1016/j.neunet.2014.06.002>
- Huang, B., Liu, Y., Lu, J., Jiang, Y., & Gui, W. (2025). Two-timescale distributed approach to Nash equilibrium seeking of Euler-Lagrange systems in multicenter non-convex games. *IEEE Transactions on Automatic Control*, 70(9), 6323–6330. <https://doi.org/10.1109/TAC.2025.3559743>
- Huang, K., Zhou, C., Qin, Y., & Tu, W. (2020). A game-theoretic approach to cross-layer security decision-making in industrial cyber-physical systems. *IEEE Transactions on Industrial Electronics*, 67(3), 2371–2379. <https://doi.org/10.1109/TIE.2019.2907451>
- Katwe, M., Singh, K., Clerckx, B., & Li, C.-P. (2023). Rate splitting multiple access for sum-rate maximization in IRS aided uplink communications. *IEEE Transactions on Wireless Communications*, 22(4), 2246–2261. <https://doi.org/10.1109/TWC.2022.3210338>
- Khalil, H. K. (2002). Nonlinear systems. (3rd ed.). Upper Saddle River: Prentice Hall.
- Kokotović, P. V. (1987). Singular perturbation techniques in control theory. In P. V. Kokotović, A. Bensoussan, & G. L. Blankenship (Eds.), *Singular perturbations and asymptotic analysis in control systems*, pp. 1–55. Berlin, Heidelberg: Springer. <https://doi.org/10.1007/BFb0007176>
- Li, G., Yan, Z., & Wang, J. (2015). A one-layer recurrent neural network for constrained nonconvex optimization. *Neural Networks*, 61, 10–21. <https://doi.org/10.1016/j.neunet.2014.09.009>
- Li, J., Li, C., Xu, Y., Dong, Z. Y., Wong, K. P., & Huang, T. (2018). Noncooperative game-based distributed charging control for plug-in electric vehicles in distribution networks. *IEEE Transactions on Industrial Informatics*, 14(1), 301–310. <https://doi.org/10.1109/TII.2016.2632761>
- Liu, F., Dong, X., Yu, J., Hua, Y., Li, Q., & Ren, Z. (2022). Distributed Nash equilibrium seeking of N-coalition noncooperative games with application to UAV swarms. *IEEE Transactions on Network Science and Engineering*, 9(4), 2392–2405. <https://doi.org/10.1109/TNSE.2022.3163447>
- Liu, J., & Yi, P. (2023). Predefined-time distributed Nash equilibrium seeking for non-cooperative games with event-triggered communication. *IEEE Transactions on Circuits and Systems II: Express Briefs*, 70(9), 3434–3438. <https://doi.org/10.1109/TCSII.2023.3259483>
- Liu, P., Xiao, F., Wei, B., & Yu, M. (2024). Generalized Nash equilibrium seeking for noncooperative games with heterogeneous individual dynamics. *IEEE Transactions on Automatic Control*, 69(4), 2492–2499. <https://doi.org/10.1109/TAC.2023.3308343>
- Lu, K., & Zhu, Q. (2022). Nonsmooth continuous-time distributed algorithms for seeking generalized Nash equilibria of noncooperative games via digraphs. *IEEE Transactions on Cybernetics*, 52(7), 6196–6206. <https://doi.org/10.1109/TCYB.2021.3049463>
- Ma, Z., Zou, S., Ran, L., Shi, X., & Hiskens, I. A. (2016). Efficient decentralized coordination of large-scale plug-in electric vehicle charging. *Automatica*, 69, 35–47. <https://doi.org/10.1016/j.automatica.2016.01.035>
- Meng, Q., Nian, X., Chen, Y., & Chen, Z. (2024). Attack-resilient distributed Nash equilibrium seeking of uncertain multiagent systems over unreliable communication networks. *IEEE Transactions on Neural Networks and Learning Systems*, 35(5), 6365–6379. <https://doi.org/10.1109/TNNLS.2022.3209313>
- Nash, J. F. (1951). Non-cooperative games. *Annals of Mathematics*, 54(2), 286–295. <https://doi.org/10.2307/1969529>
- Romano, A. R., & Pavel, L. (2020). Dynamic NE seeking for multi-integrator networked agents with disturbance rejection. *IEEE Transactions on Control of Network Systems*, 7(1), 129–139. <https://doi.org/10.1109/TCNS.2019.2920590>
- Rosen, J. B. (1965). Existence and uniqueness of equilibrium points for concave N-person games. *Econometrica: Journal of the Econometric Society*, 33(3), 520–534. <https://doi.org/10.2307/1911749>
- Salehisadaghiani, F., & Pavel, L. (2016). Distributed Nash equilibrium seeking: A gossip-based algorithm. *Automatica*, 72, 209–216. <https://doi.org/10.1016/j.automatica.2016.06.004>
- Sousa, P. R., Magalhães, L., Resende, J. S., Martins, R., & Antunes, L. (2021). Provisioning, authentication and secure communications for IoT devices on FIWARE. *Sensors*, 21(17), 5898. <https://doi.org/10.3390/s21175898>
- Stankovic, M. S., Johansson, K. H., & Stipanovic, D. M. (2012). Distributed seeking of Nash equilibria with applications to mobile sensor networks. *IEEE Transactions on Automatic Control*, 57(4), 904–919. <https://doi.org/10.1109/TAC.2011.2174678>
- Sun, C., & Hu, G. (2021). Continuous-time penalty methods for Nash equilibrium seeking of a nonsmooth generalized noncooperative game. *IEEE Transactions on Automatic Control*, 66(10), 4895–4902. <https://doi.org/10.1109/TAC.2020.3040377>
- Sun, J., & Liu, C. (2018). Distributed zero-sum differential game for multi-agent systems in strict-feedback form with input saturation and output constraint. *Neural Networks*, 106, 8–19. <https://doi.org/10.1016/j.neunet.2018.06.007>
- Tian, C., Mei, J., Tian, K., & Ma, G. (2025). Distributed adaptive Nash equilibrium seeking for games of heterogeneous high-order players over a directed graph. *Automatica*, 179, 112296. <https://doi.org/10.1016/j.automatica.2025.112296>
- Wang, H., Liu, Q., & Xu, C. (2024a). Predefined-time distributed optimization and anti-disturbance control for nonlinear multi-agent system with neural network estimator: A hierarchical framework. *Neural Networks*, 175, 106270. <https://doi.org/10.1016/j.neunet.2024.106270>

- Wang, H., Luo, H., Jiang, Y., & Wu, S. (2025). Game-based distributed decision optimization for heterogeneous multiagent systems with unknown nonlinear dynamics. *IEEE Transactions on Cybernetics*, 55(9), 4054–4063. <https://doi.org/10.1109/TCYB.2025.3581999>
- Wang, M., Chen, J., Wen, C., & Qin, S. (2024b). Adaptive generalized Nash equilibrium seeking algorithm for nonsmooth aggregative game under dynamic event-triggered mechanism. *Automatica*, 169, 111835. <https://doi.org/10.1016/j.automatica.2024.111835>
- Wang, M., Zhu, S., & Qin, S. (2023). An adaptive neurodynamic approach for solving nonsmooth N-cluster games. *Neural Networks*, 166, 595–608. <https://doi.org/10.1016/j.neunet.2023.07.041>
- Wang, X.-F., Sun, X.-M., Ye, M., & Liu, K.-Z. (2022). Robust distributed Nash equilibrium seeking for games under attacks and communication delays. *IEEE Transactions on Automatic Control*, 67(9), 4892–4899. <https://doi.org/10.1109/TAC.2022.3164984>
- Yang, H., Yang, S., Xu, Y., Cao, E., Lai, M., & Dong, Z. (2015). Electric vehicle route optimization considering time-of-use electricity price by learnable partheno-genetic algorithm. *IEEE Transactions on Smart Grid*, 6(2), 657–666. <https://doi.org/10.1109/TSG.2014.2382684>
- Ye, M., Han, Q.-L., Ding, L., & Xu, S. (2023). Distributed Nash equilibrium seeking in games with partial decision information: A survey. *Proceedings of the IEEE*, 111(2), 140–157. <https://doi.org/10.1109/JPROC.2023.3234687>
- Ye, M., & Hu, G. (2017). Distributed Nash equilibrium seeking by a consensus based approach. *IEEE Transactions on Automatic Control*, 62(9), 4811–4818. <https://doi.org/10.1109/TAC.2017.2688452>
- Ye, M., & Hu, G. (2018). Distributed Nash equilibrium seeking in multiagent games under switching communication topologies. *IEEE Transactions on Cybernetics*, 48(11), 3208–3217. <https://doi.org/10.1109/TCYB.2017.2764141>
- Ye, M., & Hu, G. (2021). Adaptive approaches for fully distributed Nash equilibrium seeking in networked games. *Automatica*, 129, 109661. <https://doi.org/10.1016/j.automatica.2021.109661>
- Yi, P., Lei, J., Li, X., Liang, S., Meng, M., & Chen, J. (2022). A survey on noncooperative games and distributed Nash equilibrium seeking over multi-agent networks. *CAAI Artificial Intelligence Research*, 1(1), 8–27. <https://doi.org/10.26599/AIR.2022.9150002>
- Yu, R., Tang, Y., Yi, P., & Li, L. (2024). Distributed Nash equilibrium seeking dynamics with discrete communication. *IEEE Transactions on Neural Networks and Learning Systems*, 35(3), 4347–4353. <https://doi.org/10.1109/TNNLS.2022.3201133>
- Zeng, Y., Hua, C., Liao, B., & Li, Z. (2025). Design and analysis of gradient-based differential neural network for solving time-varying quadratic problems with inequality constraint. *Neural Networks*, 192, 107911. <https://doi.org/10.1016/j.neunet.2025.107911>
- Zhang, K., Fang, X., Wang, D., Lv, Y., & Yu, X. (2021). Distributed Nash equilibrium seeking under event-triggered mechanism. *IEEE Transactions on Circuits and Systems II: Express Briefs*, 68(11), 3441–3445. <https://doi.org/10.1109/TCSII.2021.3068176>